NGUYEN GIA NGHI

Ho Chi Minh City, Vietnam | gianghi30032005@gmail.com | Portfolio | GitHub | LinkedIn

RESEARCH INTERESTS

Software engineering | Distributed and cloud systems | Systems performance and reliability

Debugging, observability and latency diagnosis

EDUCATION

VNUHCM-University of Science Sep 2023–May 2027 (Expected)

Bachelor of Science | Major: Information Technology Ho Chi Minh City

GPA: 8.41/10.00 (3.61/4.00) | 133 credits completed

UNDERGRADUATE RESEARCH THESIS

Research title (English translation): Latency diagnosis and optimization mechanisms for edge cloud-gaming systems

Official title (Vietnamese): Nghiên cứu cơ chế chẩn đoán và tối ưu hóa độ trễ cho hệ thống trò chơi đám mây biên

Supervisor: Professor Le Ngoc Son

Expected completion date: March 2027

- Studying whether latency fingerprints, context-specific patterns of how systems respond to a relief action, can help distinguish cross-layer bottlenecks and support more targeted recovery than generic adaptation across heterogeneous edge-cloud-gaming environments.
- Built a self-hosted edge-cloud-gaming testbed using containerized game workloads, desktop host orchestration, GStreamer capture, and WebRTC streaming. The testbed can also export cross-layer telemetry used for the research.
- Implemented the first end-to-end fingerprinting pipeline which includes versioned records, response-vector construction, fingerprint storage, and fingerprint matching. In an initial repeatability check, a separate experimental run matched the initial seed with a match strength of 0.982 across 22 shared features.
- Planning comparisons with fixed-profile, threshold-based, and passive-telemetry baselines. Comparisons will be evaluated through diagnosis accuracy, P95/P99 latency, frame-deadline misses, recovery time, quality degradation, and probe overhead. A secondary comparison will evaluate stale-work handling with and without deadline awareness.

RELEVANT ACADEMIC PREPARATION

Selected strong results (10-point scale): Discrete Mathematics (9.40) | Introduction to Cloud Computing (9.10) | Cloud Implementations and Operations (9.70) | Computer Systems (9.10) | Software Architecture (8.60) | Software Design (8.50) Applied Statistics for Engineers and Scientists (8.10) | Operating Systems (8.00) | Introduction to Databases (8.00)

SYSTEMS AND SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine

- Developed a Python pipeline and CLI that can ingest testbed data, validate records against versioned schemas, construct response vectors, and match fingerprints within compatible contexts.

Repository

PIXELATED Studio | Creator platform and self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into an edge-cloud-gaming testbed that streams interactive workloads from a host owned by users to a lightweight browser client.
- Implemented a Fastify control plane, Electron-based host orchestration, a containerized runtime, GStreamer capture and encoding, and WebRTC streaming, with cross-layer telemetry export.

Repository | Live application | Releases

PIXELATED User | Browser-based WebAssembly gaming platform

- Built and deployed a PWA application that runs NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly libretro cores, without relying on the desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export data including launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser baseline for comparison with the Studio version's edge-cloud-streaming path.

Repository | Live application

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern | TMA Solutions Jun–Aug 2026

Ho Chi Minh City

- Implemented bug fixes, product improvements, and small features for an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Worked with Mongoose data models and queries, developed and validated API tests, and contributed changes through an Agile/Scrum workflow.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Infrastructure and DevOps	Linux, Docker, Kubernetes, Terraform, GitHub Actions
Real-time media	WebRTC, GStreamer
Cloud platforms	AWS, GCP
Performance evaluation	Telemetry instrumentation, benchmarking, latency analysis

CERTIFICATIONS AND ENGLISH

- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall – Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2nd October 2025)